PHARMACOVIGILANCE INTERNSHIP

TASK 3

CAUSALITY ASSESSMENT

Detailed Case Studies & Clinical Reasoning Using the WHO-UMC Causality Framework

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Task	Causality Assessment
Assessment approach	WHO-UMC categories with supporting evidence review
Case status	Fictional educational training cases
Submission format	GitHub repository + PDF case report

Project objective: evaluate the relationship between a medicine and an adverse reaction using a structured, evidence-based approach; classify the relationship without overstating certainty; and document the reasoning in a submission-ready format.

Scope note: The Task 3 method is intentionally aligned with the concepts introduced in Task 1 and the documented case information from Task 2. The three cases demonstrate different levels of evidence under the WHO-UMC framework.

1. Learning objectives

This task is designed to demonstrate how a pharmacovigilance professional moves from a reported adverse event to a cautious causality assessment. The exercise focuses on chronology, dechallenge, rechallenge, alternative explanations, known reaction patterns, patient factors and objective evidence where available.

The central principle is that an event occurring after a medicine exposure is not automatically caused by that medicine. Causality is a reasoned assessment of the evidence actually documented in the case.

2. WHO-UMC causality categories

Category	Core WHO-UMC-style requirement
Certain / Definite	Plausible time relationship; other causes cannot adequately explain the event; clinically plausible response to withdrawal; event is pharmacologically or phenomenologically definitive; satisfactory rechallenge when necessary.
Probable / Likely	Reasonable time relationship; unlikely to be explained by concurrent disease or another drug; clinically reasonable response to withdrawal; rechallenge information is not required.
Possible	Reasonable time relationship; event could also be explained by disease or another drug; withdrawal information may be lacking or unclear.
Unlikely	Timing makes a causal relationship improbable and a disease or other drug provides a plausible alternative explanation.
Conditional / Unclassified	Reported as an adverse reaction, but more data are needed or additional information is still under examination.
Unassessable / Unclassifiable	Available information is insufficient or contradictory and cannot be supplemented or verified.

The present project concentrates on three teaching categories - Certain, Probable/Likely and Possible - because these are the three classifications explicitly highlighted by the Task 3 training brief and the supplied student sample.

3. Evidence checklist for causality assessment

Factor	Assessment question	How it is used
Temporal relationship	Did the reaction occur after exposure and within a plausible interval?	Compatible timing supports an association, but timing alone does not prove causation.
Dechallenge	Did the event improve after withdrawal?	Clinically plausible improvement can strengthen the suspected relationship.
Rechallenge	Did the event recur after re-exposure?	A positive rechallenge may provide strong supporting evidence; it is not something to create deliberately for training.
Alternative causes	Could disease, another medicine or another exposure explain the event?	A credible alternative weakens the medicine-event relationship.
Known reaction profile	Is the presentation compatible with known medicine effects?	Adds biological and clinical plausibility.
Objective evidence	Are examination, laboratory or other objective findings available?	Can strengthen event characterization when present.

4. How Task 1 and Task 2 feed into Task 3

Task 1 foundation: the earlier project introduced AE vs ADR, Type A-F patterns, temporal relationship, dechallenge, rechallenge, alternative causes, WHO-UMC, Naranjo and Bradford Hill concepts. The key lesson was to document preliminary causality carefully rather than treat chronology as proof.

Task 2 documentation: the ADR report captured the actual case fields - a 34-year-old female patient, amoxicillin 500 mg orally three times daily for acute bacterial sinusitis, treatment dates 01/03/2026-03/03/2026, generalized itching with pruritic rash beginning on 03/03/2026, recovery on 06/03/2026, withdrawal of amoxicillin, cetirizine for symptomatic relief, no rechallenge, and a preliminary Probable/Likely assessment.

Task 3 application: the case record is now interpreted more systematically against the WHO-UMC criteria. The same documented facts are used; unsupported clinical information is not invented.

5. CASE 1 - CERTAIN / DEFINITE

Case type: Fictional educational training scenario designed to demonstrate the highest evidentiary category used in this exercise.

Field	Case information
Patient	L.R., 31-year-old female, 58 kg
Indication	Acute bacterial pharyngitis
Suspect medicine	Azithromycin 500 mg orally once daily
Background medicines	None reported
Reaction	Generalized urticaria with facial swelling and wheezing
Rechallenge	Positive after inadvertent re-exposure
Outcome	Recovered after emergency treatment; suspect medicine avoided thereafter

6. Detailed chronology

Time	Exposure / event	Causality significance
Day 0 - 08:00	Azithromycin taken for acute infection.	Clear exposure anchor.
Day 0 - 08:20	Generalized urticaria, facial swelling and wheezing begin.	Very close temporal relationship.
Day 0 - 08:30	Medicine is stopped and emergency treatment is provided.	Begins dechallenge observation.
Day 0 - later	Respiratory and cardiovascular symptoms improve.	Compatible clinical improvement.
Later follow-up	The same medicine is inadvertently taken again and a similar acute reaction recurs.	Positive rechallenge evidence.

7. Initial assessment

The event is compatible with an acute hypersensitivity-type reaction. The timing is exceptionally close to exposure, no competing medicine or new exposure is documented, symptoms improve after stopping treatment, and the same pattern recurs after re-exposure.

8. CASE 1 - WHO-UMC criterion-by-criterion assessment

Criterion	Requirement	Case evidence	Met?
Time relationship	Plausible relationship to drug intake.	Reaction began approximately 20 minutes after the dose.	YES
Alternative causes	Cannot be adequately explained by concurrent disease or another drug.	No competing medicine or newly reported exposure is documented.	YES
Dechallenge	Clinically plausible response after withdrawal.	Symptoms improved after the suspect medicine was stopped and treatment was given.	YES
Rechallenge	Satisfactory re-exposure evidence when necessary.	Similar reaction recurred on inadvertent re-exposure.	YES
Definitive event pattern	Pharmacologically or phenomenologically definitive.	The reaction has a coherent immediate hypersensitivity phenotype.	YES

9. Final classification: CERTAIN / DEFINITE

This case demonstrates convergence of several independent evidence streams: exceptionally close chronology, a clinically compatible pattern, absence of an identified competing explanation, compatible improvement after withdrawal and a positive inadvertent rechallenge. Taken together, the fictional record meets the teaching criteria for Certain/Definite under the WHO-UMC framework used for this project.

10. Pharmacovigilance learning point

A positive rechallenge can be highly informative when it occurs naturally or inadvertently. However, the purpose of causality assessment is not to create a rechallenge in order to obtain a stronger classification. Patient safety takes priority; causality should be assessed from evidence that is already available.

11. Why this case is not simply "after drug = because of drug"

The classification is driven by the combination of evidence, not chronology alone. If the positive rechallenge or the lack of alternative explanations were absent, the same event would need to be considered against weaker categories.

12. CASE 2 - PROBABLE / LIKELY (LINKED TO TASK 2)

This case is intentionally carried forward from the documented Task 2 ADR report so that the internship portfolio has continuity between reporting and causality assessment.

Field	Task 2 case entry
Patient	A.K. | 34 years | Female | 62 kg
Indication	Acute bacterial sinusitis
Suspected medicine	Amoxicillin 500 mg orally three times daily
Therapy dates	01/03/2026 - 03/03/2026
Reaction	Generalized itching with pruritic rash
Onset / recovery	03/03/2026 / 06/03/2026
Action taken	Amoxicillin withdrawn; cetirizine given for symptomatic relief
Outcome / seriousness	Recovered; non-serious
Rechallenge	Not performed
Concomitant medicines	None reported
Task 2 preliminary assessment	Probable / Likely

13. Chronology reconstruction

Date	Event	Assessment relevance
01/03/2026	Amoxicillin therapy begins.	Exposure established.
03/03/2026	Generalized itching and pruritic rash occur.	Reaction follows treatment exposure and is recorded on the therapy stop date.
03/03/2026	Amoxicillin withdrawn; cetirizine used for symptomatic relief.	Dechallenge observation begins.
06/03/2026	Reaction resolves.	Supports a compatible withdrawal/recovery pattern.
After recovery	No rechallenge performed.	No re-exposure evidence is available.

14. CASE 2 - WHO-UMC criterion-by-criterion assessment

Criterion	WHO-UMC teaching requirement	Case evidence	Finding
Time relationship	Reasonable relationship to drug intake.	Reaction occurred during amoxicillin therapy and was documented on the treatment stop date.	Supports
Alternative causes	Unlikely to be explained by concurrent disease or another drug.	No concomitant medicine or new alternative exposure is reported in Task 2.	Supports
Dechallenge	Clinically reasonable response to withdrawal.	Amoxicillin was withdrawn and the reaction resolved by 06/03/2026.	Supports
Rechallenge	Not required for Probable/Likely.	No rechallenge was performed.	N/A
Known reaction profile	Presentation should be clinically compatible with the medicine.	Generalized itching/pruritic rash is compatible with a hypersensitivity-type presentation.	Supports

15. Final classification: PROBABLE / LIKELY

The documented chronology is reasonable, the reaction improves after withdrawal, and no clear competing medicine or exposure is reported. At the same time, there is no positive rechallenge and the record is limited to the fictional case information captured in Task 2. Therefore, Probable/Likely communicates meaningful evidence without overstating certainty.

16. Continuity across the internship portfolio

Task	What was demonstrated	How it contributes to Task 3
Task 1	ADR identification and causality concepts.	Provided the terminology and assessment factors.
Task 2	Structured ADR reporting of the amoxicillin case.	Provided the documented patient, medicine, reaction, timeline and outcome fields.
Task 3	Formal causality classification.	Uses the documented case evidence to assign a WHO-UMC-style category.

17. CASE 3 - POSSIBLE

Case type: Fictional training case designed to show how a compatible temporal relationship can remain weak when credible alternative explanations are present.

Field	Case information
Patient	M.P., 45-year-old male, 74 kg
Indication	Hypertension
Suspect medicine	Drug Y, fictional antihypertensive, 10 mg once daily
Concurrent circumstances	Reduced fluid intake and only 4-5 hours of sleep per night
Reaction	Mild-to-moderate headache
Dechallenge	Not evaluable - medicine continued
Rechallenge	Not performed
Outcome	Headache resolved after hydration and adequate rest while medicine continued

18. Detailed chronology

Day	Event	Causality relevance
Day 1	Drug Y started.	Exposure established.
Days 1-3	Poor sleep and reduced fluid intake continue.	Credible alternative contributors are present.
Day 3	Headache develops; Drug Y is continued.	Temporal relationship exists but is nonspecific.
Day 4	Patient rests and increases fluid intake; headache resolves later that day.	Resolution occurs without drug withdrawal.

19. Initial interpretation

The event occurs after medicine initiation, so timing is compatible. However, headache is nonspecific, dechallenge cannot be assessed, and two credible non-drug explanations are documented. Because the symptom resolves while Drug Y continues, the medicine-event relationship remains limited.

20. CASE 3 - WHO-UMC criterion-by-criterion assessment

Criterion	Requirement	Case evidence	Finding
Time relationship	Reasonable time relationship to drug intake.	Headache developed on Day 3 of treatment.	Compatible
Alternative causes	Event could also be explained by disease or another factor.	Reduced fluid intake and sleep deprivation are credible explanations.	Alternative cause present
Dechallenge	Withdrawal information may be lacking or unclear.	Drug Y was continued; withdrawal was not tested.	Not evaluable
Rechallenge	Not required for Possible.	No re-exposure was performed.	N/A
Event definitiveness	Event does not need to be definitive.	Headache is nonspecific and has multiple possible causes.	Weak support

21. Final classification: POSSIBLE

The temporal relationship supports considering the medicine as a possible contributor, but the case contains credible alternative explanations and no dechallenge evidence. Resolution while the medicine is continued further limits the causal inference. Possible therefore reflects the evidence without overstating causality.

22. Pharmacovigilance learning point

> This case demonstrates why the phrase "the symptom started after the drug" cannot by itself establish causation. A good assessor actively searches for alternative explanations and records when a criterion cannot be evaluated.

23. Documentation rule

When the case does not contain enough information to answer a causality question, document it as not available, not evaluable or unknown rather than inventing an answer.

24. Comparative interpretation

The three cases intentionally represent different evidentiary strengths. The assigned WHO-UMC category should reflect the evidence that is actually documented rather than the desired outcome of the assessment.

Evidence dimension	Case 1	Case 2	Case 3
Temporal relationship	Very strong; minutes	Reasonable; during therapy	Reasonable; Day 3
Dechallenge	Compatible improvement	Improvement after withdrawal	Not available / not evaluable
Rechallenge	Positive inadvertent recurrence	Not performed	Not performed
Alternative causes	None identified	No clear alternative reported	Credible alternatives present
Overall interpretation	Strong multi-factor evidence	Meaningful evidence with missing rechallenge	Association weakened by competing explanations
WHO-UMC teaching category	Certain / Definite	Probable / Likely	Possible

25. Practical pharmacovigilance takeaways

1. Build the chronology first: record medicine start, event onset, action taken, recovery and any re-exposure information.
2. Do not equate "after the drug" with "because of the drug".
3. Review all medicines and other exposures for alternative explanations.
4. Interpret dechallenge and rechallenge in clinical context.
5. Never invent missing information.
6. Use the WHO-UMC category that best matches the documented evidence.
7. Keep the final wording proportionate to the strength of the evidence.

26. Final quality checklist

Check	Status
Task 3 uses a structured WHO-UMC framework	COMPLETE
Three evidence-strength cases are included	COMPLETE
Task 2 amoxicillin case is carried forward without adding unsupported facts	COMPLETE
Temporal relationship, dechallenge, rechallenge and alternatives addressed	COMPLETE
Certain / Probable-Likely / Possible categories explained	COMPLETE
No deliberate re-exposure is proposed	COMPLETE
Fictional educational status clearly stated	COMPLETE

Declaration: This PDF is a fictional educational training project for a pharmacovigilance internship. It is not a real patient record, does not replace clinical assessment, and does not contain real patient or reporter identifiers.